Consider, $(-7) \times (-42)$,

$$(-7) \times (-42) = +294$$

Therefore, $(-42) \times (-7)$ and $(-7) \times (-42)$ are equal.

It is commutative.

Example 1.22

Prove that $[(-2) \times 3] \times (-4) = (-2) \times [3 \times (-4)]$.

Solution

In the first case (–2) and (3) are grouped together and in the second case (3) and (–4) are grouped together

L.H.S $= [(-2) \times 3] \times (-4)$

$= (-6) \times (-4) = 24$

R.H.S $= (-2) \times [3 \times (-4)]$

$= (-2) \times (-12) = 24$

Therefore, L.H.S. = R.H.S.

$[(-2) \times 3] \times (-4) = (-2) \times [3 \times (-4)]$ Hence it is proved.

Example 1.23

Are $(-81) \times [5 \times (-2)]$ and $[(-81) \times 5] \times (-2)$ equal? Mention the property.

Solution

Consider, $(-81) \times [5 \times (-2)]$,

$$(-81) \times [5 \times (-2)] = (-81) \times (-10) = 810$$

Consider, $[(-81) \times 5] \times (-2)$,

$$[(-81) \times 5] \times (-2) = (-405) \times (-2) = 810$$

Therefore, $(-81) \times [5 \times (-2)]$ and $[(-81) \times 5] \times (-2)$ are equal.

It is associative.

Example 1.24

Are $3 \times [(-4)+6]$ and $[3 \times (-4)]+(3 \times 6)$ equal? Mention the property.

Solution

Consider, $3 \times [(-4)+6]$,

$$3 \times [(-4)+6] = 3 \times 2 = 6$$

Consider, $[3 \times (-4)]+[3 \times 6]$,

$$[3 \times (-4)]+[3 \times 6] = -12+18 = 6$$

Therefore, $3 \times [(-4)+6]$ and $[3 \times (-4)]+3 \times 6$ are equal.

It is the distributive property of multiplication over addition.

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